# What makes the biggest difference in the LCA results?

Parameter group with the largest influence on each RAW prototype's net impact

## On its own

first-order Sobol' index: the group's effect alone, without interactions

| | Today | | | | 2050 | | | |
|---|---|---|---|---|---|---|---|---|
| | biopol | timber | cfw | knit | biopol | timber | cfw | knit |
| climate change | location | design* | design* | location | design | design* | design* | circularity |
| ozone depletion | location | design* | design* | location | manufacturing* | design | design* | location* |
| particulate matter formation | design* | design* | manufacturing* | circularity* | design | design | manufacturing* | circularity* |
| acidification | location* | location | manufacturing* | location* | design | design | manufacturing* | circularity* |
| freshwater eutrophication | location | location | design* | location | manufacturing* | design* | manufacturing* | circularity* |
| land use | manufacturing* | design* | manufacturing* | circularity* | manufacturing* | design* | manufacturing* | circularity* |
| water use | design | location | design* | location* | design | design* | design* | circularity* |
| mineral resources | manufacturing* | design* | design* | manufacturing* | manufacturing* | design* | design* | location* |

## Including interactions

total-order Sobol' index: the group's effect alone plus its interactions with other groups

| | Today | | | | 2050 | | | |
|---|---|---|---|---|---|---|---|---|
| | biopol | timber | cfw | knit | biopol | timber | cfw | knit |
| climate change | location | design | manufacturing | location | design | design | manufacturing* | circularity |
| ozone depletion | location | design | manufacturing* | location | manufacturing* | design | manufacturing* | location |
| particulate matter formation | design | design* | manufacturing | circularity | design | design | manufacturing | circularity |
| acidification | location | location | manufacturing | location | design | design | manufacturing | circularity |
| freshwater eutrophication | location | location | manufacturing | location | design | design | manufacturing | circularity |
| land use | manufacturing* | design | manufacturing* | circularity | manufacturing* | design | manufacturing | circularity |
| water use | design | location | manufacturing | location | design | design | manufacturing | circularity |
| mineral resources | manufacturing | design | manufacturing | manufacturing | location* | design* | manufacturing | location |

design: what the product is made of

manufacturing: machines, loading, yield

circularity: lifetime, repair, end of life

location: which country's electricity

* close call: the 95 % interval of the top group overlaps that of the runner-up